

Mathematics 2025/26, Homework 5
bonus: 2 points, deadline: May 11, 2026

Find the following integrals:

1. $\int (4x^5 - 3x^4 + 7x^2 + 5) \, dx$

2. $\int \frac{1}{x^8} \, dx$

3. $\int \sqrt{x} \, dx$

4. $\int x e^{2x^2+1} \, dx$

5. $\int (x + 1) \sin x \, dx$

6. $\int (5x - 4)^7 \, dx$

7. $\int \frac{1}{x+5} \, dx$

8. $\int_1^2 (x^2 + 2) \, dx$

Instructions for writing homework:

- Write your homework with solution (not only the results).
- Take a photo of the homework and convert the picture to PDF (use <https://tools.pdf24.org/en/jpg-to-pdf>). Alternatively, use a different way to make a PDF file with your homework.
- Compress the file if it is large (use <https://tools.pdf24.org/en/compress-pdf>).
- Upload the final PDF file to “cousework submissions” in UIS.